The stratum the scorecard cannot report as missing

Bias Bounty Mapping Equity Challenge — submitted for *Best Bias Discovery*. Zindi username: **[TO FILL IN]**. No leaderboard submission is associated with this writeup.

Every figure below is computed from the challenge package alone, by three scripts published with this entry. No additional data source is used, nothing is reconstructed from removed reference layers, and nothing is random: re-running returns the same tables.

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1. The discovery

The only fine-grained urban-heat layer in the challenge package is absent for 30.8% to 62.4% of tracts in every study region, and its absence is concentrated on tribal communities.

The strata tables carry a NASA UHE-Daily urban-heat-exposure block (`uhe_covered`, `uhe_city_name`, `uhe_wbgtmax28_days_per_yr`, and four more columns). It is the only measure in the package that describes *urban* heat exposure at tract level. It does not cover every tract:

region	UHE absent	rural among absent vs covered	tribal among absent vs covered	population without the stratum
`eastern-wa`	**62.4%** (259/415)	54.1% vs 0.6%	6.2% vs 0.0%	**60.4%**
`eastern-ok`	**58.0%** (691/1192)	58.2% vs 1.8%	**85.8% vs 42.3%**	**60.1%**
`northern-ca`	48.6% (287/591)	49.5% vs 1.6%	4.5% vs 0.7%	44.9%
`south-central-tx`	35.6% (2138/6010)	45.6% vs 0.4%	—	34.9%
`maricopa-az`	30.8% (490/1593)	31.6% vs 0.7%	6.3% vs 0.5%	28.9%

Five regions, one direction, rural concentration ratios of 31× to 114×.

The mechanism, demonstrated rather than assumed

The layer is **keyed by city**. `uhe_city_name` is populated for **100%** of covered tracts and **0%** of uncovered ones; `uhe_overlap_frac` has no non-null value among the uncovered. In Eastern Oklahoma the layer knows **six cities** — Tulsa, Oklahoma City, Norman, Edmond, Lawton, Fort Smith — and nothing outside them.

This is not an error by NASA. It is what an urban dataset is. The finding is that a package assembled **to measure equity** inherits a city-shaped blind spot, and that nothing in the automated scorecard reveals it.

Why the automated scorecard cannot surface this

The Bias Score computes five metrics **within** predefined strata. A tract with no value for a stratum does not produce a "missing" count — it **drops silently out of the breakdown**, and the metric returns a number for the tracts that remain.

So the scorecard can report "the coverage gap is larger among rural tracts". It cannot report "for 58% of this region, 85.8% of them tribal, this stratum does not exist". The heat disparity it publishes is computed over a population that systematically excludes the communities the challenge exists to protect.

And the pattern does not fit the predefined strata

Most of the largest affected tracts are classified `Urban` by RUCA. A tract can be urban and still fall outside the six city polygons the layer knows.

The absence therefore **does not follow rurality — it follows membership of one of six named cities**. An urban/rural split cannot express this, because *both sides* of that cut contain uncovered tracts. The stratum that explains the pattern is one the API does not have: *is this tract inside a UHE city?*

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2. The evidence

Eastern Oklahoma — the Cherokee / Muscogee / Choctaw region, which the challenge itself labels "Drought + **heat**" — carries the finding.

tracts scored	1 192
without the urban-heat stratum	**691 (58.0%)**
of those, tribal	**593**
population in uncovered tracts	**2 360 128**
tribal areas affected	**26**, totalling **2 013 570 residents**

By tribal area (full list in `resultats/tracts-sans-uhe.csv`):

area	tracts	population	median SVI
Cherokee OTSA	131	456 814	0.658
Creek OTSA	103	334 494	0.626
Chickasaw OTSA	93	304 578	0.635
Choctaw OTSA	65	216 800	0.743
Cheyenne and Arapaho OTSA	55	202 874	0.523
Kiowa-Comanche-Apache-Fort Sill Apache OTSA	37	118 594	0.589
Citizen Potawatomi Nation-Absentee Shawnee OTSA	30	113 753	0.532
Kaw/Ponca joint-use OTSA	6	24 095	**0.823**
Seminole OTSA	8	19 724	**0.798**

Across all 26 affected areas, **9 have a median SVI above 0.75** — the most socially vulnerable quartile — and **25 of 26 above 0.5**. The heat stratum is missing almost everywhere social vulnerability sits above the national median.

Named tracts, so judges reopen exactly these:

GEOID	tribal area	population	SVI	RUCA class
40143005809	Cherokee OTSA	8 362	0.296	Urban
40143006709	Creek OTSA	8 244	0.280	Urban
40131050404	Cherokee OTSA	7 678	0.258	Urban
40009966200	Cheyenne and Arapaho OTSA	7 078	**0.880**	Urban
40143005805	Cherokee OTSA	7 369	**0.808**	Urban
40027202405	Citizen Potawatomi Nation-Absentee Shawnee	6 617	0.707	Rural

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3. Who is affected, and what it changes

Urban heat exposure is what decides **where cooling centres go**, **which neighbourhoods get a heat alert first**, and **who is doorknocked before a forecast event**. Those decisions are made at tract level, from exactly the kind of layer that is missing here. In the terms this challenge uses: the absence bears on **emergency dispatch** — which tracts a heat-illness surge is anticipated in, and where crews are staged before it — and on **disaster relief** — where a cooling centre, a water distribution point or a wellness check is placed once an event is under way. It bears less on evacuation routing, which heat rarely triggers, and we do not claim it does.

A tract with no UHE value cannot be ranked for heat risk by any analysis built on this package. It is not ranked low — it is **absent from the ranking**, which is worse, because an absent tract produces no number for anyone to question.

The consequence is specific to these places. Eastern Oklahoma's tribal statistical areas are designated drought-prone and heat-exposed by the challenge's own framing, and they are where the layer is most often missing: **2.0 million people across 26 tribal areas**, including 456 814 in the Cherokee OTSA alone. Tract `40009966200` (Cheyenne and Arapaho OTSA, 7 078 residents, SVI 0.880) sits in the most vulnerable 12% nationally and has no urban-heat value at all.

This is the challenge's own hypothesis, one layer further in: *"Where those gaps overlap with wildfire corridors, extreme-heat cities, and drought-prone tribal lands, the consequences can be life-threatening."* The gap we report is not in the map — it is in the instrument used to decide who the map is failing.

A negative result we checked, and report

We tested whether uncovered tracts are *also* the hottest, which would have made a direct life-safety case. **Pooled across the five regions, they are not.** Using EPHT summer heat-days over the 9 792 scored tracts that carry one — of 9 801; nine tracts in South-Central Texas have no EPHT value — 19.6% of tracts without the UHE stratum fall in the top exposure quartile, against 31.0% of covered ones, a

ratio of **0.63×**. The intersection of "no UHE stratum" and "top heat quartile" is 754 tracts, of which only 3 are tribal.

That pooled quartile mostly sorts climates rather than tracts, and we say so. South-Central Texas and Maricopa are hotter in absolute terms than Oklahoma, Northern California and Eastern Washington: three of the five regions have **no tract at all** above the common threshold. Recomputing the quartile **within each region**, which separates the absence from the climate, the direction is not uniform:

region	local threshold (summer heat-days)	top quartile, no UHE	top quartile, with UHE	ratio
`south-central-tx`	16.6	12.1%	32.5%	0.37×
`eastern-ok`	11.8	19.8%	34.3%	0.58×
`maricopa-az`	15.3	25.3%	33.4%	0.76×
`eastern-wa`	11.9	32.8%	16.7%	**1.97×**
`northern-ca`	11.6	53.3%	8.2%	**6.48×**

In Eastern Washington and Northern California the missing stratum falls **disproportionately on the locally hottest tracts**. We publish this because an entry reporting only the pooled figure would be overturned by a judge changing one line, and because it is the stronger reading of our own finding in two of the five regions.

The pooled mechanism is nonetheless real: EPHT heat-days run higher in cities because of the urban heat island, and the UHE layer covers cities. That is why the two measures disagree — and why neither can stand in for the other.

This narrows the claim, and we state it narrowed. The finding is not that these communities are hotter: pooled they are cooler, and within region it depends on the region. It is that they **cannot be assessed on urban heat at all** — any analysis reaching for the package's only urban-heat measure silently drops 58% of Eastern Oklahoma, 85.8% of that tribal, and reports a disparity computed on the remainder. An unmeasurable place is not a safe place; it is a place no number argues for.

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4. What the scorecard should add

Three changes, each a column or a count rather than a new dataset:

1. **Publish stratum coverage beside every metric.** Each of the five metrics should report the number of scored tracts it was computed over, and the number dropped for want of a stratum value. A metric that silently loses 58% of a region reads identically to one that loses none.
2. **Add an `is_uhe_city` boolean.** The pattern we report is invisible to the urban/rural split because *both sides* contain uncovered tracts. One boolean — is this tract inside a UHE city polygon — would expose it inside the existing hazard-exposure breakdown instead of hiding it.
3. **Flag city-keyed layers as such in the data dictionary.** UHE-Daily is indexed by city by construction; a user who does not unnest `uhe_city_name` has no way to know its coverage is a function of municipal boundaries rather than of data quality.

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5. Reproducing this

Everything below is at **https://github.com/Yoh5/bias-bounty-uhe** — the three scripts, the result tables they produce, and this write-up.

Three scripts, provided data only, no randomness:

```
python src/couverture_strates.py   # coverage of every stratum, all 5 regions
python src/tracts_nommes.py        # the named tracts and tribal areas
python src/quartile_chaleur.py     # the negative result of section 3, both thresholds
```

The first downloads `strata/<region>/<region>-strata-tract-table.parquet` from the public Source Cooperative repository and caches it; the other two read that cache. Outputs land in `resultats/couverture-strates.csv`, `resultats/tracts-sans-uhe.csv` and `resultats/quartile-chaleur.csv`.

Three practical notes that cost us time:

- **Read `GEOID` as text.** Maricopa is FIPS 04 and an integer read eats the leading zero, breaking every join. The challenge warns of this; it is worth repeating because the failure is silent.
- **The repository returns 403 to urllib's default user agent** while the S3 listing endpoint does not. An explicit `User-Agent` header fixes it.
- **South-Central Texas carries 6 010 tracts, not the 6 003 the README announces.** The 6 010 GEOIDs are distinct, so this is not duplication. Every figure here is computed over the tables as delivered, and our scripts print the discrepancy rather than silently adopting either count.

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6. What we did not establish

We could not confirm which heat layer the Bias Score API consumes. The challenge page names "NOAA/CDC HeatRisk"; the repository README documents CDC EPHT and NASA GEHE / UHE-Daily, and does not specify the API's strata. If the API's heat stratum is drawn from EPHT — which is complete across all five regions except nine tracts in South-Central Texas — then the finding is that the *complete but coarse* layer conceals a hole in the *fine-grained* one, rather than that the API's own stratum is blind. That is a weaker claim than the strongest reading, and we state it rather than assume the stronger one.

Either way the operative point survives: any analyst reaching for urban-heat exposure in this package inherits a city-shaped absence that falls on tribal and non-city communities, and no metric in the scorecard reports it.

We did not cross this with Overture coverage. Showing that the blind spot falls where mapping is already weakest would strengthen the impact argument, and it requires the 0.9 GB reference package for Eastern Oklahoma. We report what we verified.

Two pattern families were deliberately set aside because the organisers published them, and restating an organiser's own finding is not a discovery: the cross-region incomparability of the transport component (Overture/TIGER ratio 0.71 to 1.59) and the exclusion of hospitals (Overture at roughly 12× the reference).